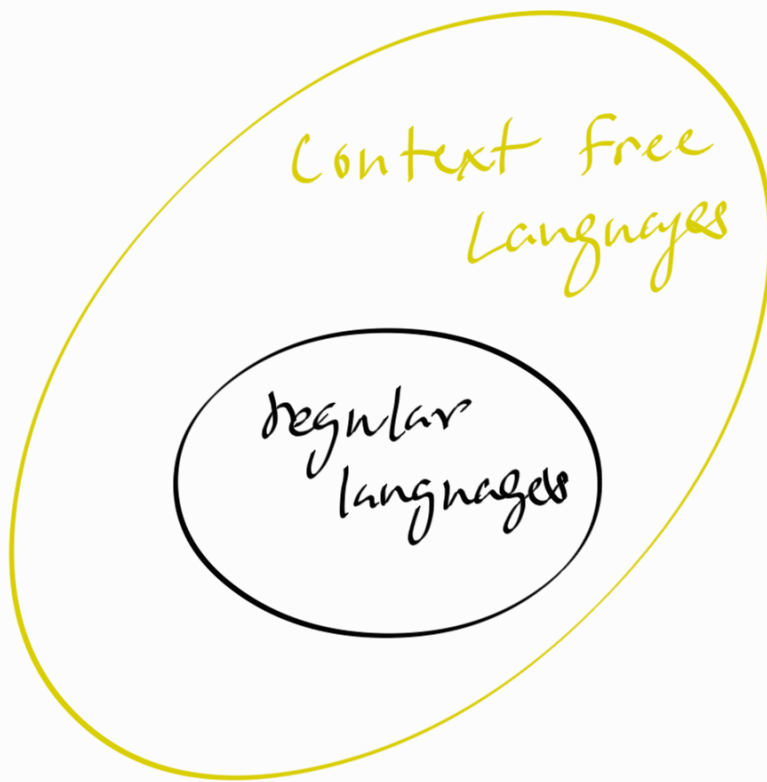


Pumping Lemma for CFL

* Every regular language is a context free language.

— NFA is basically a PDA without the stack.



Now, we are going to look at how to show a language is not context free.

For CFLs, we have slightly different pumping lemma.

Example for non-CFL

$$A = \{ a^n b^n c^n \mid n \geq 0 \}$$

not context free

First we will look at the pumping Lemma.

Pumping Lemma for CFLs.

If A is a context free language, then there is a number p (the pumping length) where, If s is a string in A of length at least p , then s may be divided into five pieces $s = uvxyz$, satisfying the conditions

1. $uv^ixy^iz \in A, \forall i \geq 0$

2. $|v| > 0$

3. $|vxy| \leq p$

← either v or y has to be a non-empty string

Proof idea:

A is a CFL.

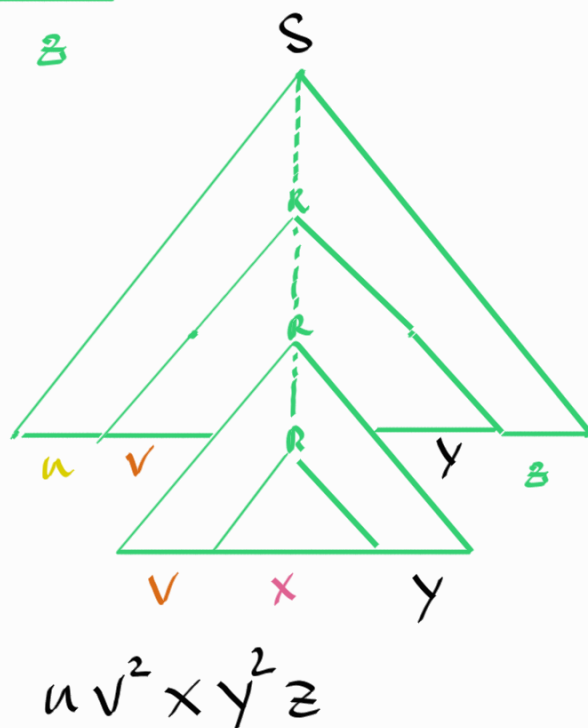
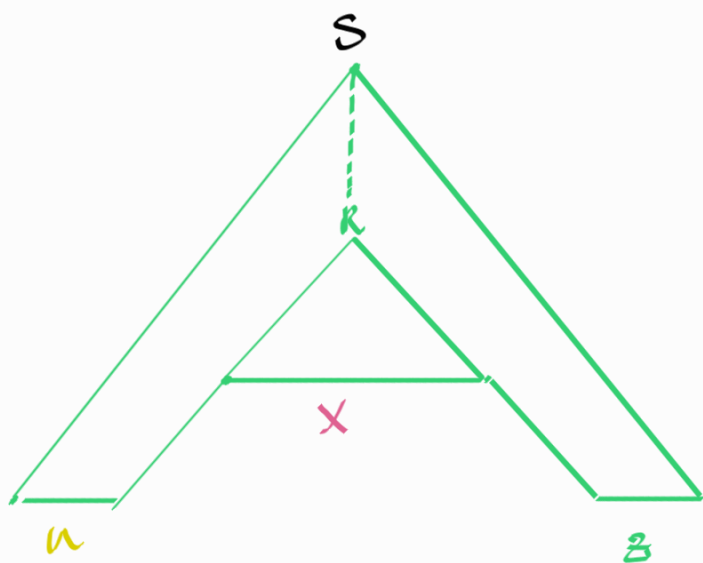
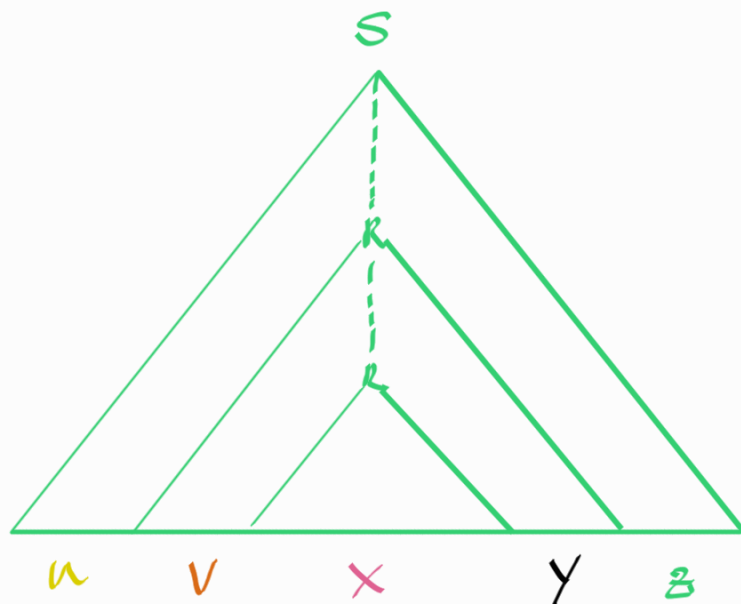
we pick a large string S in A .

G is the grammar of A .

$$S \rightarrow \underline{XY} \mid _ \mid$$

$$X \rightarrow _ \mid _ \mid$$

$$Y \rightarrow _ \mid _$$



Proof:

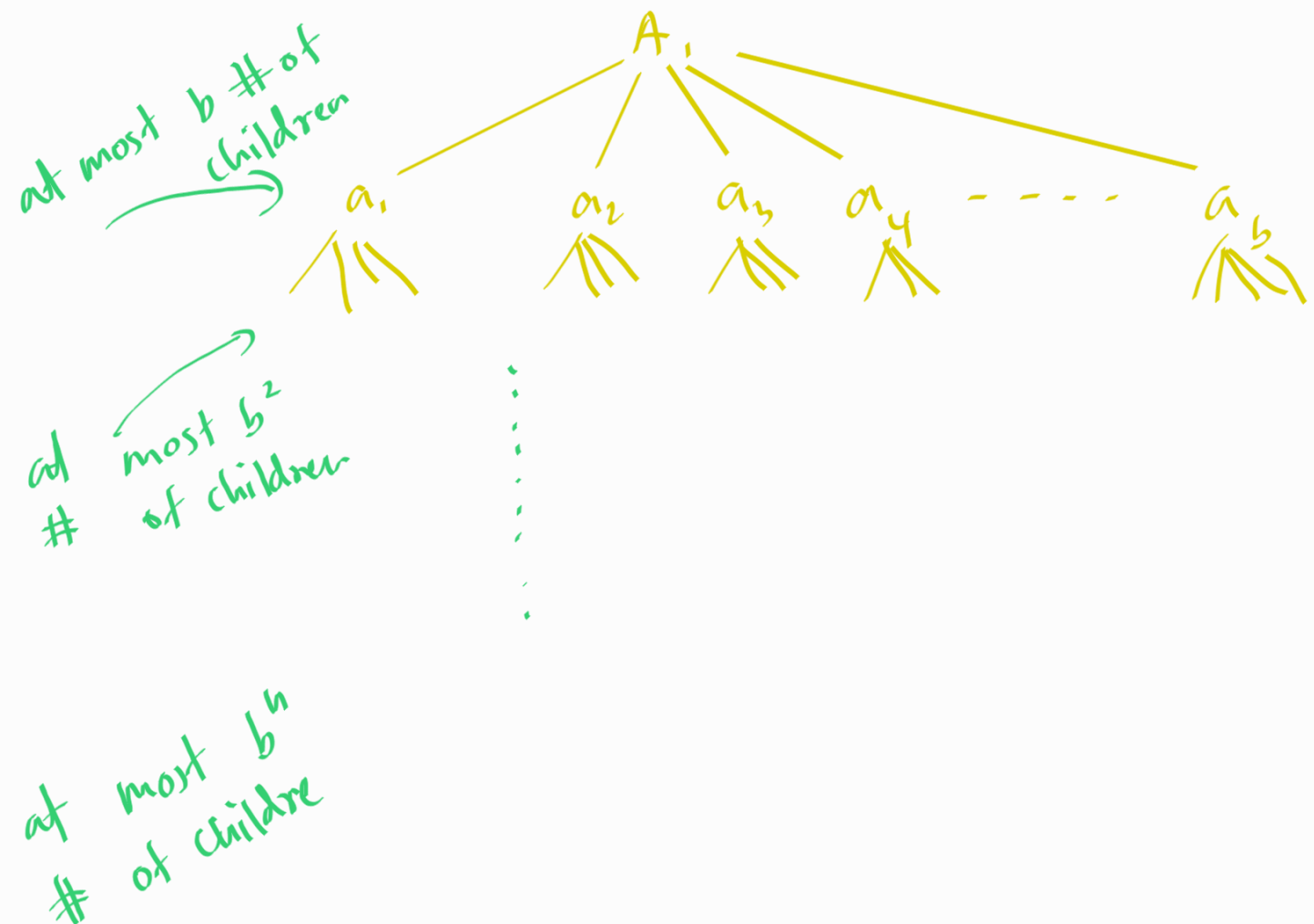
Suppose A is CFL.

Suppose max size of any rule in grammar G that describes A is b .

$S \rightarrow$

$A_1 \rightarrow a_1 a_2 \dots a_b$

This means in the worst case, we can use this rule, and in the parse tree we could have b children



If you use h steps in the generation process, then the length of the string that you can generate is at most b^h .

If the string s is length $b^h + 1$.

Let the # of variables in G be $|V|$.

Let $p = b^{|V|+1}$

Suppose s is a string in A s.t. $|s| \geq p$.

Let $\bar{T}$ be one of the parse trees of s . Pick the parse tree that uses the minimum number of variables.

Pick the longest path in $\bar{T}$ from root to leaf.

and it must be at least $|V|+1$.

The # of nodes in this path must be at least $|V|+2$.

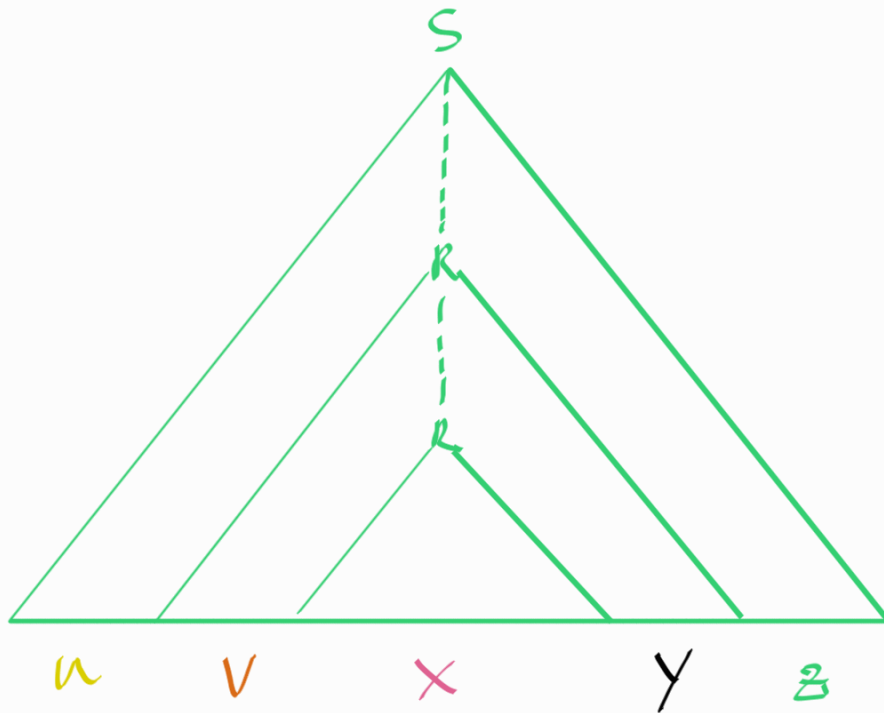


This includes one at terminal, and $|V|+1$ variable nodes.

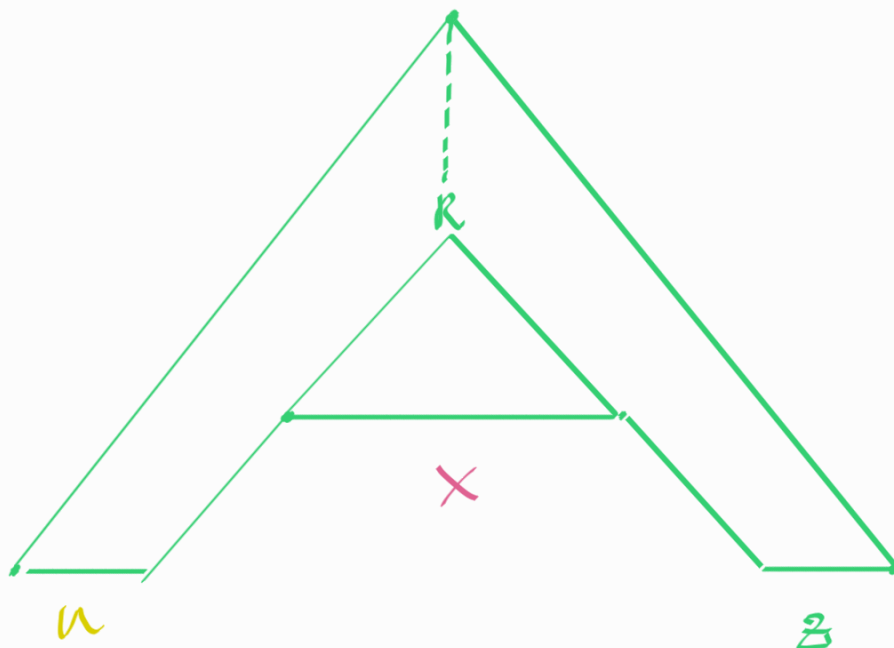
$\therefore$ we must have reused at least one variable twice.

Let R be one of the variables that is repeated and among the lowest $|V|+1$ variables on the path.

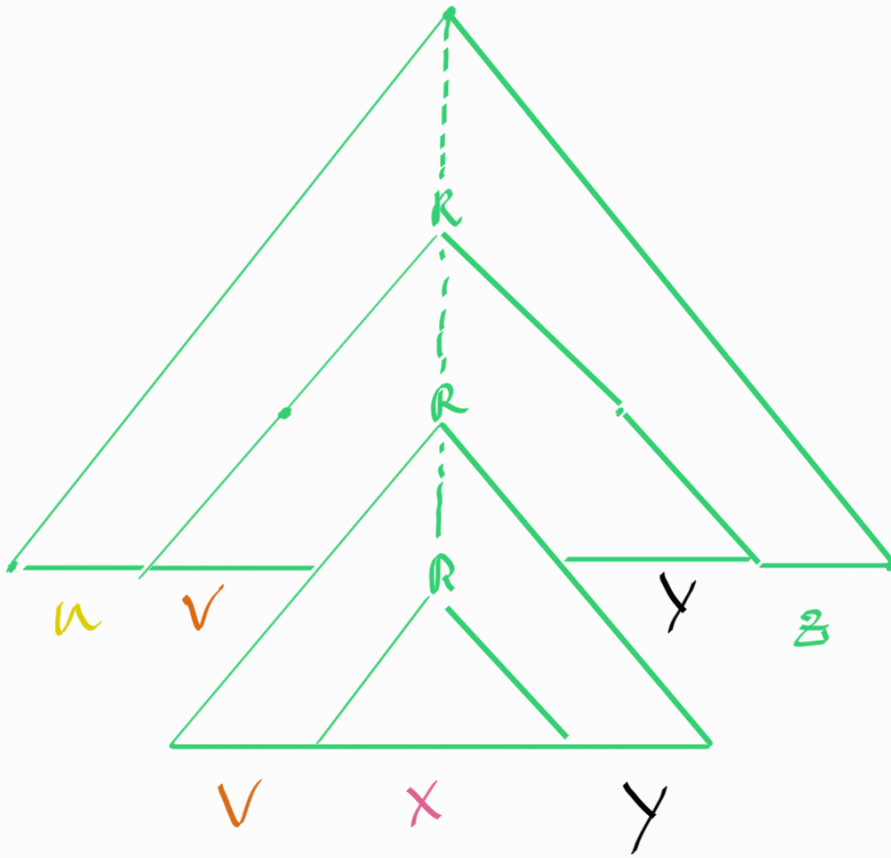
Using the previous figure, we can decompose S into $uxyz$.



Replacing larger subtree R with smaller subtree creates the string uxz .



Replacing smaller subtree R with larger subtree R i # of times creates the string $uv^i x y^i z$.


$$uv^2 \times y^2 z$$

$$vv^3 \times y^3 z$$

This establishes condition ① in the lemma.

To get the condition 2, we must show that both v and y cannot be ε .

Note, that if both v and y are ε ,
then the parse tree obtained for
 $u x z$ must contain fewer nodes
than the original parse tree for $u \varepsilon x \varepsilon z$
 $= u x z$.

But this cannot happen as we picked the smallest possible parse tree for s .

$\therefore$ condition 2 is satisfied.

For condition 3, we need to show that $|vxy| \leq p$.

Note that we picked R to be a variable in the bottom $|V|+1$ variables in the parse tree.

Thus, height of the larger subtree R is $|V|+1$.

Height of this tree can only create $(|V|+1)$ a string of length at most $b^{(|V|+1)} = p$.

Hence, condition 3 is satisfied.

$\therefore$ we conclude that pumping lemma holds.

